Rupal Tripathi

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Skills

- **Languages**: C, C++, Python, SQL
- **Libraries**: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
- **Tools/Platforms**: Excel, Power BI, Jupyter Notebook, VS Code, Google Colab, Git, GitHub, FastAPI, Docker
- **Soft Skills**: Problem Solving, Communication, Public Speaking, Team Collaboration, Time Management

Projects

Deepfake Audio Authenticity Detection System | GitHub — Jan' 26

- Built an end-to-end system to classify real vs AI-generated speech, including an ML pipeline, FastAPI backend, and simple web interface.
- Designed an audio preprocessing and feature engineering pipeline extracting 48 features (MFCCs, RMS energy, spectral centroid, zero-crossing rate) using Python and Librosa.
- Deployed the FastAPI prediction service on Render with containerized configuration, enabling real-time audio authenticity predictions.

Tech Stack: Python, Librosa, NumPy, Scikit-learn, FastAPI, Docker, HTML, CSS, JavaScript

Startup Funding Analytics Dashboard | GitHub — Sept' 25

- Cleaned and transformed Indian startup funding data and designed an optimized data model for analytical reporting in Power BI.
- Created DAX measures to compute KPIs such as total funding, funding trends, and top investor contributions across sectors and cities.
- Delivered a multi-page interactive dashboard with drill-downs, filters, and geospatial views to support exploratory and insight-driven analysis.

Tech Stack: Power BI, Power Query, DAX, Excel

AgriPredict : Climate-Aware Crop Recommendation Engine| GitHub — Jul' 25

- Developed a machine learning-based crop recommendation system that suggests suitable crops based on soil parameters and climate conditions.
- Integrated a live weather API for real-time crop prediction and trained models, and built an interactive interface with visualization and insights.
- Achieved ~95% prediction accuracy and deployed the application on Streamlit Cloud with downloadable recommendation reports.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Weather API, Streamlit, Matplotlib

Certificates

- Cloud Computing : NPTEL | — Apr' 25
- Data Structures and Algorithms : iamneo (72 hours) | — Dec' 24
- Computer Communications Specialization : University of Colorado (Coursera)| — Nov' 24
- Introduction to Relational Databases : IBM, Coursera | — Mar'24
- Responsive Web Design Developer Certification : freeCodeCamp | — Nov' 23

Achievements

- **GHCI Hackathon 2025:** — Nov' 25
  Competed in a National Hackathon focused on problem-solving and solution design under time and evaluation constraints.
- **Finalist - SAP Hackfest 2025:** — Aug' 25
  Selected among the **Top 70 teams out of 5,000+** Nationwide for developing an Ethical AI Governance Toolkit.
- **Naukri CodeQuest#23 Rank Holder:** — Jul' 25
  Secured **Rank 251 among 18,788+ participants** in a National-level competitive coding challenge organized by Naukri Campus.

Education

Lovely Professional University — Punjab, India
Bachelor of Technology - Computer Science and Engineering; CGPA: 7.5 — Aug' 23 - Present

Woodland Academy — Gorakhpur, Uttar Pradesh
Intermediate; Percentage: 77% — Apr' 21 - Mar' 22

Little Flower School — Gorakhpur, Uttar Pradesh
Matriculation; Percentage: 95% — Apr' 19 - Mar' 20